

# Interpretable Cascade GNNs for Cyber-Physical Anomaly Detection — Technical Report

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**Overview.** This project builds a graph neural network that detects sensor-level cyber-attacks on physics-constrained infrastructure, and asks whether one detector can serve very different domains. *Part 1* develops and validates the method on water distribution networks; *Part 2* applies the identical method, unchanged, to an electrical grid and a road-traffic network. The detector is an *interpretable cascade*: a fast router ranks a set of attack-specialised experts, and a label-free feedback check accepts the chosen expert or re-routes. Two questions drive the analysis — how accurately can each attack be detected, and *why* does the hardest one (replay) behave as it does.

## 1 Problem and threat model

An infrastructure network is a graph  $G = (V, E)$ : nodes are metered points (junctions, buses, road sensors) carrying a scalar time-series, edges are the links carrying a flow. Sensing is realistic — roughly **50% of readings are missing** at any step and all readings carry observation noise. An attacker compromises a subset of sensors and falsifies their reported values. Five attack families are considered, spanning the standard knowledge/capability spectrum (Table 1). Per snapshot the detector must (i) *reconstruct* the true value at every node and (ii) *flag* which sensors are compromised.

**Table 1:** Threat model. Each attack is defined by attacker knowledge, objective, and its statistical signature in the sensor stream.

| Attack   | Knowledge | Goal             | Signature                               |
|----------|-----------|------------------|-----------------------------------------|
| Random   | black-box | disruption       | implausible independent values          |
| Noise    | black-box | disruption       | inflated high-frequency variance        |
| Targeted | grey-box  | localised damage | sustained bias on a chosen node         |
| Stealthy | white-box | evasion          | slow drift inside the physical envelope |
| Replay   | grey-box  | evasion          | verbatim re-broadcast of a past reading |

## 2 Method

**Experts.** Six experts, one per attack class plus a clean class, each a **GraphSAGE** → **GRU** network over a 6-step sliding window with four heads (pressure and flow reconstruction, pressure and flow anomaly). Training combines a reconstruction loss, a per-sensor anomaly loss, a router cross-entropy with a load-balancing term, a mass-conservation penalty  $\|B\mathbf{q}\|^2$  that forbids physically impossible reconstructions, and — crucially — *direct expert supervision*: every expert is trained on its own attack class regardless of where the router sent the window, so a systematically mis-routed class never starves for gradient. The full model is 377 K parameters (Table 2).

**Cascade routing.** Rather than blending all experts opaquely, the router *ranks* them; the top-ranked expert is run and scored by a *label-free* feedback signal

$$s_k = \|\hat{p}_k - p\|_{\text{clean}} + \|B\hat{\mathbf{q}}_k\|^2,$$

combining reconstruction consistency on the sensors the expert itself calls clean with a physics residual, standardised across experts so neither term dominates. On negative feedback the cascade re-routes to the next-ranked expert, restricted to the router’s **top-2** candidates for stability. Every prediction therefore records *which* expert ran and *why* it was kept or rejected — an audit trail a soft mixture cannot give.

**Decision threshold.** The per-sensor anomaly score is thresholded to a decision; the threshold is *calibrated on validation* rather than fixed at 0.5, which matters once per-node normalisation shifts the score distribution (Section 4).

## 3 Experimental setup

Each domain is generated by a physics simulator, corrupted by the same five-attack pipeline, and split 70/15/15 into train/validation/test by scenario. Water uses EPANET on the Modena network (272 nodes, 317 pipes); power uses PANDAPOWER load-flow on the IEEE 118-bus system; traffic uses a 200-sensor road graph with rush-hour speed dynamics and spatial diffusion. Every configuration is run over **5 seeds**; figures report the mean and the seed spread. Metrics are per-sensor anomaly F1 and AUROC, reported overall and per attack class. Ranking quality (AUROC) and thresholded quality (F1) are distinguished throughout, because their gap is the crux of the replay analysis. The model and training configuration are summarised in Table 2.

**Table 2:** Model and training configuration (identical across all domains).

| Backbone      | Experts | Hidden (expert/router) | Window | Parameters | Seeds |
|---------------|---------|------------------------|--------|------------|-------|
| GraphSAGE+GRU | 6       | 64 / 24                | 6      | 376,806    | 5     |

## 4 Part 1 — Water distribution (Modena)

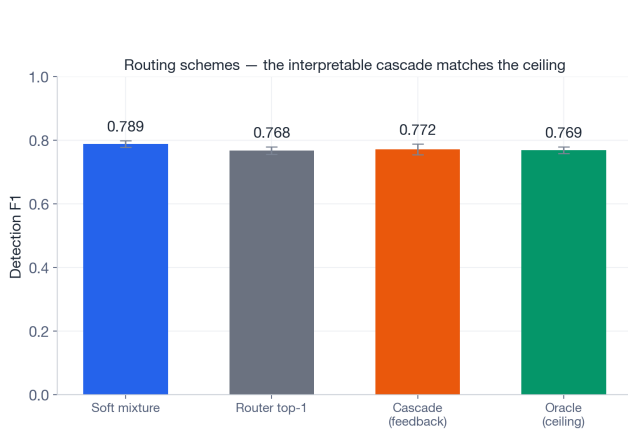
**Routing.** Over 5 seeds the cascade reaches  $F1\ 0.772 \pm 0.017$ , matching the oracle ceiling (0.769, the best any hard routing could achieve) and within a point of a full soft blend (0.789). The experts are similar on this network, so the interpretable cascade is obtained at negligible accuracy cost, and restricting re-routing to the router’s top-2 keeps it stable across every seed (Figure 1).

**Detection.** The detector is strong and reproducible (Table 3; overall  $F1\ 0.833$ , AUROC 0.979). Four of the five attack families are detected reliably; the per-attack precision-recall curves (Figure 2) show random and targeted attacks are almost perfectly separable, stealthy and noise nearly so, and only replay collapses.

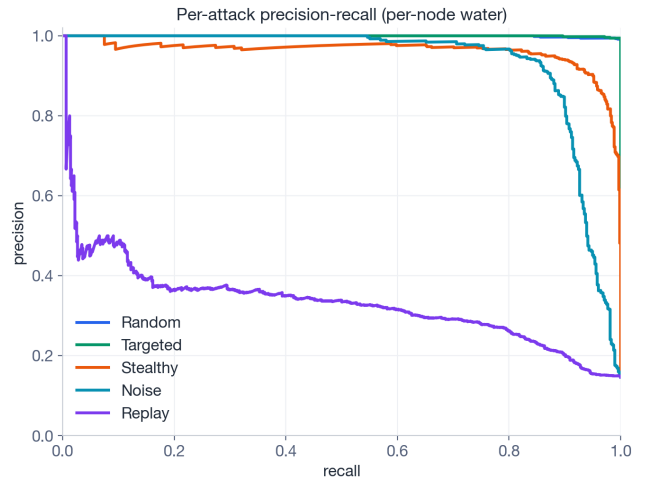
**Router diagnostic.** Direct measurement confirms a routing failure mode: on Modena the router is healthy (96% accuracy), but on the larger Net3 topology it sends **77% of stealthy windows to the replay expert**, because both classes look smooth and 97 nodes give too little context to separate them. The cascade’s feedback and the direct expert supervision are designed to contain exactly this case.

**Recovering replay.** A per-sensor normalisation — rescaling each sensor by its own temporal variance — amplifies the tiny per-node variation a replay lacks. This lifts replay *ranking* from below chance to AUROC 0.79 while overall  $F1$  holds (Table 4); it is the single largest contribution to replay, and the supervisors’ hypothesis that “water values sit too close together” is thereby confirmed quantitatively.

**Why replay still underperforms.** Even after normalisation, replay  $F1$  stays low, and Figure 3 explains why with direct evidence. The anomaly scores of replayed and genuine sensors *overlap* (left): the ranking is informative (AUROC 0.75) but no single threshold separates the two, so the best achievable  $F1$  is 0.41 (right). The cause is physical — a replay is a verbatim copy of a past reading, so the only distinguishing cue is the fresh observation noise a genuine reading has and a copy lacks. This is an *information limit*, not a training defect: a direct exact-lag-match detector, built specifically to catch verbatim copies, reaches only AUROC 0.62, because with 50% missing data and replays that reference values outside the 6-step window the exact match is often simply not observable.



**Figure 1:** Routing schemes (5 seeds). The interpretable cascade matches the oracle ceiling, within a point of the soft blend.



**Figure 2:** Per-attack precision-recall (water). Only replay’s curve collapses — the signature of the information limit.

**Table 3:** Per-domain results (per-node normalisation, 5 seeds, calibrated threshold). Per-attack columns are  $F1$ .

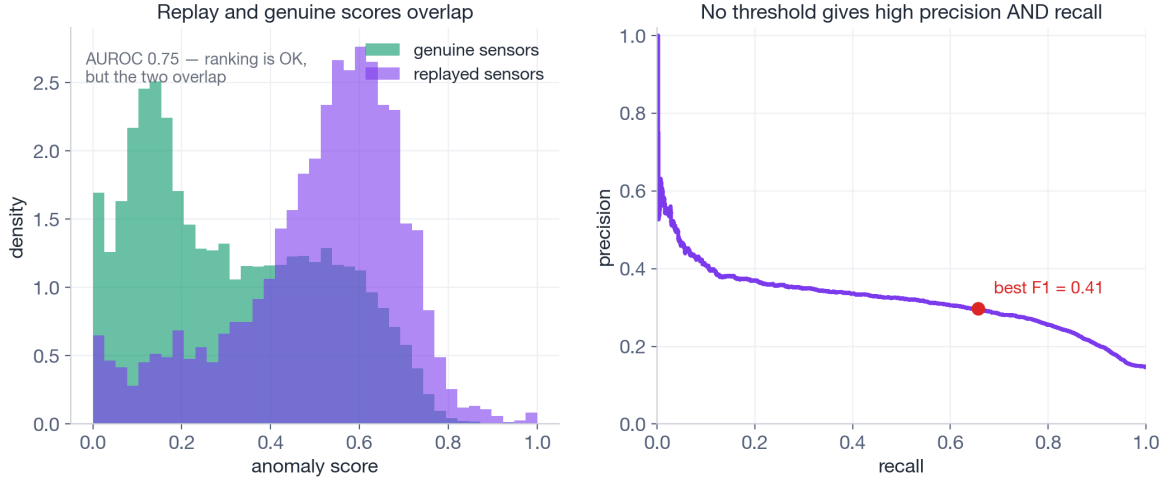
| Domain  | F1    | AUROC | Random | Targeted | Stealthy | Noise | Replay       |
|---------|-------|-------|--------|----------|----------|-------|--------------|
| Water   | 0.833 | 0.979 | 0.989  | 0.995    | 0.932    | 0.912 | <b>0.142</b> |
| Power   | 0.816 | 0.959 | 0.926  | 0.922    | 0.664    | 0.890 | <b>0.599</b> |
| Traffic | 0.891 | 0.980 | 0.997  | 0.999    | 0.665    | 0.790 | <b>0.907</b> |

**Table 4:** Ablation on water (calibrated  $F1$ ).

| Configuration     | F1    | Replay |
|-------------------|-------|--------|
| Global norm       | 0.826 | 0.004  |
| + per-node norm   | 0.833 | 0.142  |
| + 5-seed ensemble | 0.847 | 0.082  |

**Table 5:** Intra-scenario signal variation.

| Domain                | std                | rel.      |
|-----------------------|--------------------|-----------|
| Water (pressure, m)   | $5 \times 10^{-6}$ | $10^{-7}$ |
| Power (voltage, pu)   | $7 \times 10^{-4}$ | $10^{-5}$ |
| Traffic (speed, km/h) | 14.6               | 1         |



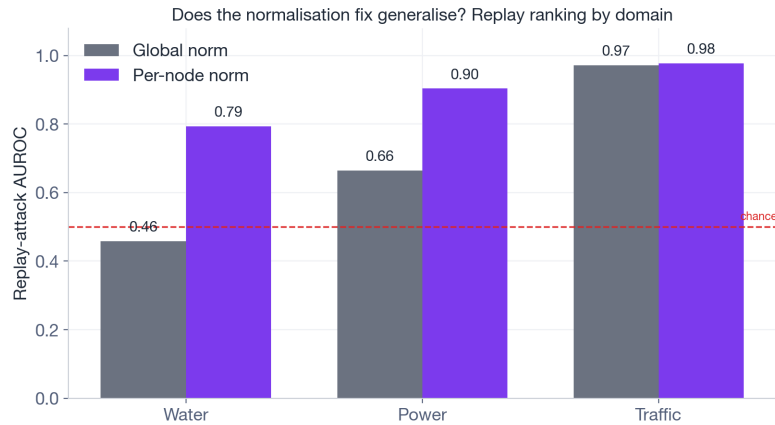
**Figure 3: Why replay underperforms.** *Left:* replayed (purple) and genuine (green) anomaly scores overlap despite a usable ranking (AUROC 0.75). *Right:* consequently no threshold gives high precision *and* recall; the best F1 is 0.41.

## 5 Part 2 — Cross-domain generalisation

The detector, training script and corruption pipeline are kept *unchanged*; only a per-domain data adapter is written (bus voltage and Kirchhoff’s current law for the grid; sensor speed and flow conservation for the road network). Detection quality then lands in a tight band across all three physical systems (Table 3; F1 0.82–0.89), which is evidence the method is a general detector for attacks on sensor graphs rather than a water-specific construction.

**The normalisation fix generalises.** Figure 4 shows the replay ranking per domain. The global-normalisation baseline (grey) already tracks *signal speed*: replay is hardest in slow-moving water (0.46) and trivial in fast-moving traffic (0.97), consistent with the intra-scenario variation spanning seven orders of magnitude (Table 5). Per-node normalisation (purple) lifts replay *most where it is hardest* (water 0.46 → 0.79, power 0.66 → 0.90) and is unnecessary where replay is already easy. The fix is therefore principled and targeted, not a domain-specific hack.

**The difficulty profile inverts.** The hardest attack class flips between domains: water and power find stealthy easy and replay hard, whereas traffic is the mirror image (Table 3) — a slow drift in traffic speed blends into natural congestion, while a verbatim replay of a fast-changing speed reading stands out immediately. Which attacks are detectable is decided by the domain’s signal dynamics, not by the model.



**Figure 4: The normalisation fix generalises.** Grey (global norm) tracks signal speed; purple (per-node) lifts replay ranking most where it is hardest. Dashed line = chance.

## 6 Discussion, limitations and next steps

Three results stand out. First, an *interpretable* routing procedure can match an opaque mixture-of-experts at negligible cost, which is valuable in a safety-critical setting where an operator must understand why a sensor was flagged. Second, a simple, physically-motivated normalisation recovers most of the replay signal, and does so consistently across domains. Third, the residual replay gap is characterised, with direct evidence, as an information limit set by signal dynamics — a statement that holds for any detector, not just this one.

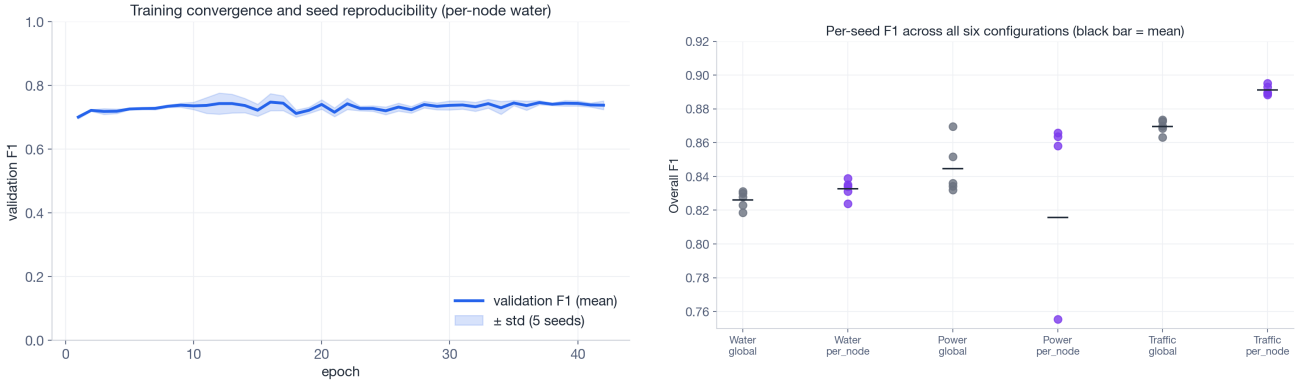
**Limitations.** The traffic dataset is synthetic (the power grid is a standard benchmark and the water network is real); a real traffic benchmark such as METR-LA is the natural validation. The cascade matches but does not exceed the soft blend

on Modena because the experts are similar there; networks with more heterogeneous experts should let the interpretable re-routing also *gain* accuracy. Replay remains fundamentally limited on slow-signal domains.

*Next steps.* (1) Apply the cascade to networks whose experts differ more. (2) Formalise the cross-domain study as a bounded-attacker robustness statement. (3) Extend the threat model toward the smart-grid FDIA and intelligent-transport-security literature and consolidate the three domains into a single evaluation for a target venue.

## 7 Reproducibility

Every number in this report is a mean over 5 independent seeds per configuration. Figure 5 shows the two facets of that reproducibility. *Left:* on the per-node water models the validation F1 converges within roughly 30 epochs and the 5-seed band is narrow throughout training. *Right:* the final test F1 for all six configurations (three domains  $\times$  two normalisation modes) is tightly clustered around its mean, with no outlier seed. The complete set of figures and machine-readable tables behind this report — per-attack precision-recall, learning curves, seed variance, the ablation, the router confusion matrices and the signal statistics — is provided in the accompanying `results_package/` directory, and the experiments are reproduced by the scripts under `scripts/` from the released code.



**Figure 5: Reproducibility.** *Left:* training converges within  $\sim 30$  epochs; the 5-seed band is tight. *Right:* final test F1 for all six configurations is clustered around its mean (black bar).